

§ 2 电磁场的普遍规律

§ 2.1 电荷守恒定律

1. 电荷密度: $\rho := \epsilon_{00} \frac{\partial \phi}{\partial t}$, $\sigma := \epsilon_{00} \frac{\partial \phi}{\partial s}$, $\lambda := \epsilon_{00} \frac{\partial \phi}{\partial l}$

$$Q = \int_V dV \rho(\vec{r}) / \int_S dS' \sigma(\vec{r}') / \int_V dV' \lambda(\vec{r}')$$

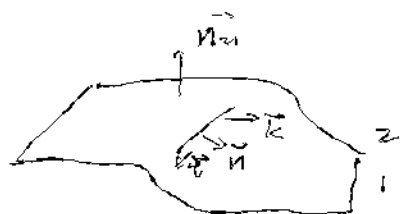
点电荷模型: $\rho(\vec{r}) = \sum_i q_i \delta(\vec{r} - \vec{r}_i)$

2. 电流 $I = \frac{dQ}{dt} = \int_S d\vec{S} \cdot \vec{j}$

电流密度 $\vec{j} = \rho \vec{v}$

3. 面电流密度: $\vec{K} = \sigma \vec{v}$

$$\begin{aligned} dI &= d(\vec{K} \cdot \vec{n}) \\ &= d(\vec{K} \cdot (\vec{n}_1 \times \vec{e})) = \underline{d\vec{e}} \cdot (\vec{K} \times \vec{n}_1) \\ &= \vec{K} \cdot \vec{n}_1 \times d\vec{l} \end{aligned}$$



4. 电荷守恒

$$q_s d\vec{S} \cdot \vec{j} = - \frac{dQ}{dt} = - \frac{d}{dt} \int_V \rho dV$$

$$\hookrightarrow \int_V dV (\vec{v} \cdot \vec{j} + \frac{d\rho}{dt}) = 0$$

稳恒电流: $\vec{v} \cdot \vec{j} = 0$

§ 2.2 电荷与电场

1. Coulomb 定律

$$\vec{F} = \frac{1}{4\pi\epsilon_0} \sum_i q_i Q_i \frac{\vec{R} - \vec{R}_i}{|\vec{R} - \vec{R}_i|^3} = q \cdot \frac{1}{4\pi\epsilon_0} \sum_i Q_i \frac{\vec{R} - \vec{R}_i}{|\vec{R} - \vec{R}_i|^3}$$

$$2. \vec{F}(\vec{R}) = q \vec{E}(\vec{R})$$

$$\vec{E}(\vec{R}) = \frac{1}{4\pi\epsilon_0} \sum_i Q_i \frac{\vec{R} - \vec{R}_i}{|\vec{R} - \vec{R}_i|^3}$$

连续情形:

$$\vec{E}(\vec{R}) = \frac{1}{4\pi\epsilon_0} \int_V dV' \rho(\vec{R}') \frac{\vec{R}}{r^3} \quad r = \vec{R} - \vec{R}'$$

3. 静电场的散度

$$* \nabla r = e_i \partial_i \sqrt{x_j x_j} = e_i \partial_i \sqrt{x_i x_i}$$

$$\nabla \frac{1}{r} = -\frac{1}{r^2} \nabla r = -\frac{\vec{r}}{r^3}$$

$$= e_i \frac{1}{r^2} \frac{x_i}{\sqrt{x_j x_j}} = e_i \frac{x_i}{r^3} = \frac{\vec{r}}{r^3}$$

$$\nabla^2 \frac{1}{r} = \nabla \cdot \left(-\frac{\vec{r}}{r^3}\right) = -4\pi\delta(\vec{r})$$

$$\nabla \cdot \vec{E}(\vec{R}) = \frac{1}{\epsilon_0} \rho(\vec{R})$$

$$\oint_S d\vec{S} \cdot \vec{E}(\vec{R}) = \int_V dV \frac{1}{\epsilon_0} \rho(\vec{R})$$

4. 静电场的旋度

$$\vec{E}(\vec{R}) = -\frac{1}{4\pi\epsilon_0} \int_V dV' \rho(\vec{R}') \nabla \frac{1}{r}$$

$$\nabla \times \vec{E}(\vec{R}) = -\frac{1}{4\pi\epsilon_0} \int_V dV' \rho(\vec{R}') (\nabla \times \nabla \frac{1}{r}) = 0$$

5. 点电荷极子

$$\varphi(\vec{R}) = \frac{1}{4\pi\epsilon_0} \frac{q}{R}$$

$$\varphi_P = \oint_{\partial V} (\varphi_+ + \varphi_-)$$

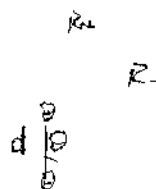
$$= \oint_{\partial V} \frac{1}{4\pi\epsilon_0} \frac{P}{d} \left(\frac{1}{R_+} - \frac{1}{R_-} \right)$$

$$= \oint_{\partial V} \frac{1}{4\pi\epsilon_0} \frac{P}{d} \left(R - d \cos\theta - R + \frac{d^2}{2R} \cos^2\theta \right)$$

$$= \frac{1}{4\pi\epsilon_0} \frac{P}{d} \frac{d \cos\theta}{R^2} = \frac{\vec{P} \cdot \vec{R}}{4\pi\epsilon_0 R^3}$$

$$\vec{E}_P = -\nabla \varphi_P = -\frac{1}{4\pi\epsilon_0} \nabla \frac{\vec{P} \cdot \vec{R}}{R^3} = -\frac{1}{4\pi\epsilon_0} \left[\frac{\nabla(\vec{P} \cdot \vec{R})}{R^3} - \frac{3(\vec{P} \cdot \vec{R})}{R^4} \nabla R \right]$$

$$= \frac{1}{4\pi\epsilon_0} \left[3(\vec{P} \cdot \vec{R}) \frac{\vec{R}}{R^4} - \frac{\vec{P}}{R^3} \right]$$



§ 2.3 电流与磁场

1. Biot-Savart 定律

$$\vec{B} = \frac{\mu_0}{4\pi} \oint_L \vec{I} d\vec{l} \times \left(\oint_V \vec{I}' d\vec{l}' \times \frac{\vec{r}}{r^3} \right)$$

2. Biot-Savart 定律

$$\vec{B} = \oint_L \vec{I} d\vec{l} \times \vec{B}$$

$$\vec{B} = \frac{\mu_0}{4\pi} \oint_L \vec{I} d\vec{l} \times \frac{\vec{r}}{r^3} = \frac{\mu_0}{4\pi} \int_V \vec{J}(\vec{r}') \times \frac{\vec{r}}{r^3} dV', \quad \vec{r} = \vec{R} - \vec{r}'$$

匀速运动的点电荷激发磁场:

$$\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi} \frac{q\vec{v} \times \vec{r}}{r^3} = \frac{e_0 \mu_0}{4\pi} \vec{v} \times \vec{E}$$

3. 磁场的散度

$$\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi} \int_V \vec{J}(\vec{r}') \cdot \vec{r} \times \frac{\vec{r}}{r^3} dV' \quad \nabla \times (\varphi \vec{A}) = (\nabla \times \vec{A}) \varphi + \nabla \varphi \times \vec{A}$$

$$= \frac{\mu_0}{4\pi} \int_V \left\{ \nabla \times \left[\frac{\vec{J}(\vec{r}')}{r} \right] - \left[\nabla \times \vec{J}(\vec{r}') \right] \frac{1}{r} \right\} dV'$$

$$= \frac{\mu_0}{4\pi} \int_V \nabla \times \left[\frac{\vec{J}(\vec{r}')}{r} \right] dV' = \nabla \times \vec{B}$$

$$\nabla \cdot \vec{B} = \nabla \cdot (\nabla \times \vec{A}) = 0$$

磁矢势

$$\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi} \int_V \frac{\vec{J}(\vec{r}')}{r} dV'$$

4. 磁场的旋度

$$\nabla \times \vec{B} = \nabla \times (\nabla \times \vec{A}) = \nabla (\nabla \cdot \vec{A}) - \nabla^2 \vec{A}$$

$$\nabla \cdot \vec{B} = \frac{\mu_0}{4\pi} \int_V \nabla \cdot \left[\frac{\vec{J}(\vec{r}')}{r} \right] dV' = \frac{\mu_0}{4\pi} \int_V \vec{J}(\vec{r}') \cdot \nabla \left(\frac{1}{r} \right) dV'$$

$$= - \frac{\mu_0}{4\pi} \int_V \vec{J}(\vec{r}') \cdot \frac{\vec{r}}{r^3} dV'$$

$$= - \frac{\mu_0}{4\pi} \int_V \left\{ \nabla' \cdot \left[\frac{\vec{J}(\vec{r}')}{r} \right] - \frac{1}{r} \nabla' \cdot \vec{J}(\vec{r}') \right\} dV' \quad \text{恒定电流 } \nabla' \cdot \vec{J}(\vec{r}') = 0$$

$$= - \frac{\mu_0}{4\pi} \int_V \vec{J}(\vec{r}') \cdot \nabla' \left(\frac{1}{r} \right) dV'$$

$$= - \frac{\mu_0}{4\pi} \oint_{S'} \vec{J}(\vec{r}') \cdot \frac{\vec{r}}{r^3} dS' = 0$$

Coulomb 规范:

$$\vec{J}(\vec{r}') \cdot \vec{r} = 0, \quad \nabla \cdot \vec{B}(\vec{r}) = 0$$

$$\begin{aligned}
 \nabla \cdot \vec{A} &= \frac{\mu_0}{4\pi} \int_V \alpha V' \cdot \nabla \cdot \frac{\vec{J}(\vec{R}')}{r} = \frac{\mu_0}{4\pi} \int_V \alpha V' \nabla \cdot \vec{R} \nabla \cdot \frac{1}{R} \\
 &= \frac{\mu_0}{4\pi} \int_V \alpha V' \vec{J}(\vec{R}') \cdot [-4\pi \delta(\vec{r})] \quad \int_V \alpha V' \delta(\vec{R}-\vec{R}') = 1 \\
 &= -\mu_0 \vec{J}(\vec{R})
 \end{aligned}$$

$$\Rightarrow \nabla \times \vec{B}(\vec{R}) = \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A} = \mu_0 \vec{J}(\vec{R})$$

$$\oint \vec{A} \cdot d\vec{l} = \mu_0 \oint \alpha \vec{J} \cdot d\vec{l} \quad \vec{J}(\vec{R})$$

5. 5.2 环状电流模型

$$\vec{A}(\vec{R}) = \frac{\mu_0}{4\pi} \oint \frac{d\vec{l}'}{r}$$

$$\vec{A}_m = \frac{\mu_0}{4\pi} \frac{\vec{m} \times \vec{R}}{R^3}$$

$$\begin{aligned}
 \vec{B} &= \nabla \times \vec{A}_m = \frac{\mu_0}{4\pi} \nabla \times \frac{\vec{m} \times \vec{R}}{R^3} > \nabla \times (\vec{f} \times \vec{g}) = (\vec{f} \cdot \nabla) \vec{g} + (\vec{g} \cdot \nabla) \vec{f} - (\vec{f} \cdot \nabla) \vec{g} - (\vec{g} \cdot \nabla) \vec{f} \\
 &= \frac{\mu_0}{4\pi} \left[(\vec{R} \cdot \nabla) \vec{m} + (\vec{m} \cdot \nabla) \vec{R} - (\vec{m} \cdot \nabla) \frac{\vec{R}}{R^3} - (\vec{R} \cdot \nabla) \frac{\vec{R}}{R^3} \right] \\
 &= \frac{\mu_0}{4\pi} \left[(\vec{R} \cdot \nabla) \vec{m} - (\vec{m} \cdot \nabla) \frac{\vec{R}}{R^3} \right] \quad \vec{R} \cdot \frac{\vec{R}}{R^3} = \frac{\vec{R} \cdot \vec{R}}{R^3} = \frac{R^2}{R^3} = \frac{1}{R} \quad \vec{R} \cdot \frac{\vec{R}}{R^3} = \frac{1}{R} \\
 &= -\frac{\mu_0}{4\pi} (\vec{m} \cdot \nabla) \frac{\vec{R}}{R^3} \\
 &= \frac{\mu_0}{4\pi} \frac{1}{R^3} \left[3(\vec{m} \cdot \vec{R}) \vec{R} - \vec{m} \right]
 \end{aligned}$$

§ 2.4. Maxwell 方程组

1. Faraday 定律

$$\mathcal{E} = - \frac{d\Phi_B}{dt}, \quad \mathcal{E} = \int_L d\vec{l} \cdot \vec{E}, \quad \Phi_B = \int_S d\vec{S} \cdot \vec{B}$$

$$\hookrightarrow \nabla \times \vec{E} = - \frac{\partial \vec{B}}{\partial t}$$

2. 位移电流

$$\text{非稳恒电流: } \nabla \cdot \vec{J} = \frac{\partial}{\partial t} \epsilon_0 (\nabla \cdot \vec{E}) = \nabla \cdot (\vec{J} + \frac{\partial}{\partial t} \epsilon_0 \vec{E}) = 0$$

$$\nabla \times \vec{B} = \mu_0 (\vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t})$$

$$\text{满足 } \nabla \cdot (\nabla \times \vec{B}) = 0 = \mu_0 \nabla \cdot (\vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t}) = 0$$

$$\text{位移电流: } \vec{J}_D = \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

3.

$$\begin{cases} \nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0} \\ \nabla \times \vec{B} = \mu_0 (\vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t}) \\ \nabla \cdot \vec{B} = 0 \\ \nabla \times \vec{E} = - \frac{\partial \vec{B}}{\partial t} \end{cases}$$

4. 电磁波波动方程

$$\text{真空中: } \begin{cases} \nabla \cdot \vec{E} = 0 \\ \nabla \times \vec{B} = \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \\ \nabla \cdot \vec{B} = 0 \\ \nabla \times \vec{E} = - \frac{\partial \vec{B}}{\partial t} \end{cases}$$

$$\rightarrow \begin{cases} (\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}) \vec{E} = 0 \\ (\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}) \vec{B} = 0 \\ \nabla \cdot \vec{E} = 0 \\ \nabla \cdot \vec{B} = 0 \end{cases} \quad \text{其中 } c = \frac{1}{\sqrt{\epsilon_0 \mu_0}}$$

5. Lorentz 力密度

$$\vec{F} = q\vec{E} + q\vec{v} \times \vec{B}$$

$$\begin{aligned} d\vec{F} &= dV \vec{f}(\vec{r}, t) = dV \left[\rho(\vec{r}, t) \vec{E}(\vec{r}, t) + \rho(\vec{r}, t) \vec{v} \times \vec{B}(\vec{r}, t) \right] \\ &= dV \left[\rho(\vec{r}, t) \vec{E}(\vec{r}, t) + \vec{j}(\vec{r}, t) \times \vec{B}(\vec{r}, t) \right] \end{aligned}$$

$$\hookrightarrow \vec{f} = \rho \vec{E} + \vec{j} \times \vec{B}$$

§ 2.5 介质中的 Maxwell 方程组

1. 介质 (被极化/极化) 产生新的电荷 / 电流附加项.

$$\rho = \rho_f + \rho_p$$

$$\vec{J} = \vec{J}_f + \vec{J}_p + \vec{J}_m$$

2. 介质极化和极化强度.

$$\vec{P} = \frac{1}{\epsilon_0} \frac{\partial \vec{p}}{\partial V} = n \vec{p} = n q \vec{r}$$

3. 极化电荷密度

$$\oint_S d\vec{S} \cdot n q \vec{E} = \oint_S d\vec{S} \cdot \vec{P} = - \Delta q_f$$

$$\hookrightarrow \oint_S d\vec{S} \cdot \vec{P} + \int_V dV \rho_p = \int_V (\vec{D} \cdot \vec{P} + \rho_p) dV = 0$$

$$\vec{D} \cdot \vec{P} + \rho_p = 0$$

在介质表面, 极化电荷面密度为 σ_p

$$\text{对电中性介质: } \oint_S d\vec{S} \cdot \vec{P} + \int_V dV \rho_p = 0$$

$$\oint_S d\vec{S} \cdot \vec{P} = \oint_S d\vec{S} (\vec{n} \cdot \vec{P}) = \int_V dV (\vec{D} \cdot \vec{P}) = - \int_V dV \rho_p$$

$$\hookrightarrow \sigma_p = \vec{P} \cdot \vec{n}$$

4. 孤立介质受到的电场力

$$\vec{F} = \int_V dV \vec{E} \rho = \int_V dV \vec{E} (\rho_f + \rho_p) + \oint_S d\vec{S} \vec{E} \sigma_p$$

$$= \int_V dV (\rho_f - \vec{D} \cdot \vec{P}) \vec{E} + \oint_S d\vec{S} \cdot \vec{P} \vec{E}$$

$$= \int_V dV [(\rho_f - \vec{D} \cdot \vec{P}) \vec{E} + \vec{D} \cdot (\vec{P} \vec{E})]$$

$$= \int_V dV [(\rho_f - \vec{D} \cdot \vec{P}) \vec{E} + (\vec{D} \cdot \vec{P}) \vec{E} + \vec{P} (\vec{D} \cdot \vec{E})]$$

$$= \int_V dV (\rho_f + \vec{P} \cdot \vec{D}) \vec{E} dV$$

5. 介质中无自由电荷, $\rho_f = 0$.

$$\vec{F} = \int_V dV (\vec{P} \cdot \vec{D}) \vec{E} \quad \text{存在不均匀的电场中受力}$$

介质只占很小 $\rho_f = 0$, 均匀极化.

$$\vec{F} \approx \int_V \rho_p \cdot \vec{v} \cdot \vec{E} = \langle \vec{p} \cdot \vec{v} \rangle \vec{E} = \langle \vec{p} \cdot \vec{v} \rangle \vec{E}$$

例: 证明点电荷 q 作用在介质颗粒上的电场力为 $\vec{F} = -k \frac{q^2}{R^3} \vec{e}_R$

$$\vec{p} = \alpha \epsilon_0 \vec{E}, \quad \vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{R^2} \vec{e}_R$$

$$\vec{F} = \alpha \epsilon_0 \langle \vec{E} \cdot \vec{v} \rangle \vec{E} = \frac{\alpha q^2}{16\pi^2 \epsilon_0} \left(\frac{\vec{e}_R \cdot \vec{v}}{R^2} \right) \frac{\vec{e}_R}{R^2}$$

$$\frac{\partial}{\partial x} \left(\frac{x}{R^3} \right) = \frac{1}{R^3} (R^3 - 3x^2 \frac{x}{R}) = \frac{1}{R^3} (R^3 - 3x^2)$$

$$\frac{\partial}{\partial y} \left(\frac{x}{R^3} \right) = \frac{(-3x^2)}{R^4} \frac{y}{R}, \quad \frac{\partial}{\partial z} \left(\frac{x}{R^3} \right) = \frac{-3x^2}{R^4} \frac{z}{R}$$

$$= \frac{\alpha q^2}{16\pi^2 \epsilon_0} \left[\frac{x}{R^3} \frac{1}{R^3} (R^3 - 3x^2) + \frac{y}{R^3} \frac{1}{R^3} (-3x^2 y) + \frac{z}{R^3} \frac{1}{R^3} (-3x^2 z) \right]$$

$$= \frac{\alpha q^2}{16\pi^2 \epsilon_0} \left(\frac{x}{R^6} - 3 \frac{x^3}{R^6} \right)$$

$$= -\frac{\alpha q^2}{8\pi^2 \epsilon_0} (x, y, z) = -k \frac{q^2}{R^3} \vec{e}_R$$

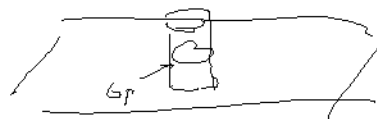
5. 介质分界面上极化电荷密度

$$\text{分界面: } \int_{\partial V} d\vec{s} \cdot \epsilon_p + q_s \int_{\partial V} d\vec{s} \cdot \vec{p} = 0$$

例: $\rightarrow 0$

$$= \int_{\partial V} (\vec{p}_2 \cdot \vec{n} - \vec{p}_1 \cdot \vec{n}) d\vec{s} + \int_{\partial V} \vec{p} \cdot d\vec{s}$$

$$\epsilon_p + (\vec{p}_2 - \vec{p}_1) \cdot \vec{n} = 0$$



6. 介质中的 Maxwell 方程

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0} = \frac{1}{\epsilon_0} (\rho_f + \rho_p) \quad \nabla \cdot \vec{p} = \rho_p$$

$$\nabla \cdot (\epsilon_0 \vec{E} + \vec{p}) = \rho_f$$

$$\vec{D} = \epsilon_0 \vec{E} + \vec{p}$$

$$\nabla \cdot \vec{D} = \rho_f$$

7. 极化电流密度

$$\text{极化电荷守恒: } \nabla \cdot \vec{J}_p + \frac{\partial \rho_p}{\partial t} = 0 \quad \rho_p = \nabla \cdot \vec{p}$$

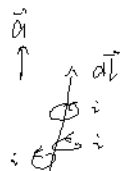
$$\nabla \cdot \vec{J}_p = -\frac{\partial \rho_p}{\partial t}$$

8. 磁化和磁化电流密度

$$\vec{M} = \frac{1}{\Delta V} \sum \vec{m}_i$$

$$d\vec{I}_m = \int \vec{m}_i \cdot d\vec{l} = n \vec{m}_i \cdot d\vec{l} = \vec{M} \cdot d\vec{l}$$

$$|\vec{dl}| = ds, \quad i\vec{dl} = \vec{m}$$



$$\hookrightarrow I_m = \oint_L \vec{m} \cdot d\vec{l} = \oint_S \vec{J}_m \cdot d\vec{S}$$

$$\vec{J}_m = \nabla \times \vec{M}, \quad \nabla \cdot \vec{J}_m = 0 \Rightarrow \text{磁化电流不会产生电荷.}$$

9. 介质中的 Maxwell 方程组.

$$\begin{aligned} \nabla \times \vec{B} &= \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \\ &= \mu_0 (\vec{J}_f + \vec{J}_p + \vec{J}_m) + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \end{aligned}$$

$$\hookrightarrow \nabla \times (\vec{B} - \mu_0 \vec{M}) = \mu_0 \vec{J}_f + \mu_0 (\epsilon_0 \frac{\partial \vec{E}}{\partial t} + \vec{J}_p)$$

$$\nabla \times \vec{H} = \vec{J}_f + \frac{\partial \vec{D}}{\partial t}$$

$$\begin{cases} \nabla \cdot \vec{D} = \rho_f \\ \nabla \times \vec{H} = \vec{J}_f + \frac{\partial \vec{D}}{\partial t} \\ \nabla \cdot \vec{B} = 0 \\ \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \end{cases}$$

10. 介质的电磁响应特性

(1) 电介质

$$\text{各向同性: } \vec{D} = \chi \epsilon_0 \vec{E}, \quad \vec{D} = \vec{P} + \epsilon_0 \vec{E} = \epsilon \vec{E}, \quad \epsilon = \chi + 1 \epsilon_0$$

$$\text{各向异性: } D_j = \epsilon_{ij} E_i$$

$$\text{非线性: } D_i = \epsilon_{ij} E_j + \epsilon_{ijk} E_j E_k + \dots$$

(2) 磁介质

$$\vec{M} = \chi_m \vec{H}, \quad \vec{B} = \vec{M} + \mu_0 \vec{H} = \mu \vec{H}, \quad \mu = 1 + \chi_m$$

(3) Ohm 定律.

$$\vec{J} = \sigma \vec{E}$$

例: 证明在均匀介质内部, 极化电荷密度 ρ_p , 自由电荷密度 ρ_f 的关系为

$$\rho_p = \left(\frac{\epsilon_0}{\epsilon} - 1\right) \rho_f$$

$$\begin{aligned} \text{证: } \rho_p &= -\nabla \cdot \vec{P} = -\nabla \cdot (\chi \epsilon_0 \vec{E}) \\ &= -\nabla \cdot \left(\frac{\chi}{\mu \chi} \vec{B}\right) = -\frac{\epsilon - \epsilon_0}{\epsilon} \rho \\ &= \left(\frac{\epsilon_0}{\epsilon} - 1\right) \rho \end{aligned}$$

$$\hookrightarrow \text{在均匀介质内部的电荷. } q' = \left(\frac{\epsilon_0}{\epsilon} - 1\right) q = \frac{\epsilon_0}{\epsilon} q$$

电偶极子: $\vec{p}' = \frac{\epsilon_0}{\epsilon} \vec{p}$

§ 2.6 介质分界面电磁场边值关系

$$\begin{cases} \oint_S d\vec{S} \cdot \vec{D} = \int_V dv \rho_f \\ \oint_L d\vec{l} \cdot \vec{H} = \int_S d\vec{S} \cdot \vec{J}_f + \frac{d}{dt} \int_S d\vec{S} \cdot \vec{D} \\ \oint_S d\vec{S} \cdot \vec{B} = 0 \\ \oint_L d\vec{l} \cdot \vec{E} = -\frac{d}{dt} \int_S d\vec{S} \cdot \vec{B} \end{cases}$$

1. 法向分量

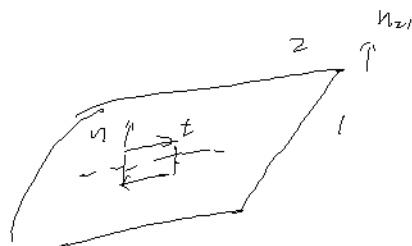
$$\begin{cases} (\vec{D}_2 - \vec{D}_1) \cdot \vec{n}_{21} dS = q_f dS \rightarrow D_{2n} - D_{1n} = \sigma_f \\ (\vec{B}_2 - \vec{B}_1) \cdot \vec{n}_{21} dS = 0 \rightarrow B_{2n} = B_{1n} \end{cases}$$

$$\begin{cases} 0 \cdot \vec{J}_f + \frac{\partial q_f}{\partial t} = 0 \rightarrow J_{f2n} - J_{f1n} = \sigma_f \\ 0 \cdot \vec{J}_p + \frac{\partial q_p}{\partial t} = 0 \rightarrow J_{p2n} - J_{p1n} = \sigma_p \\ 0 \cdot \vec{J}_m = 0 \rightarrow J_{m2n} = J_{m1n} \end{cases}$$



2. 切向分量

$$\begin{cases} \oint_L d\vec{l} \cdot \vec{H} = \int_S d\vec{S} \cdot \vec{J}_f + \frac{d}{dt} \int_S d\vec{S} \cdot \vec{D} \rightarrow \vec{n}_{21} \times (\vec{H}_2 - \vec{H}_1) = \vec{K}_f \\ \oint_L d\vec{l} \cdot \vec{E} = -\frac{d}{dt} \int_S d\vec{S} \cdot \vec{B} \rightarrow \vec{n}_{21} \times (\vec{E}_2 - \vec{E}_1) = 0 \end{cases}$$



$$\begin{aligned} (\vec{H}_2 - \vec{H}_1) \cdot \vec{t} dl &= K_f \cdot \vec{n} dl = K_f dl \cdot (\vec{n}_{21} \times \vec{t}) \\ &= (K_f \times \vec{n}_{21}) \cdot \vec{t} dl \end{aligned}$$

$$\begin{aligned} \vec{n}_{21} \times (\vec{H}_2 - \vec{H}_1) \cdot \vec{t} &= \vec{n}_{21} \times (K_f \times \vec{n}_{21}) \cdot \vec{t} \\ &= [K_f (\vec{n}_{21} \cdot \vec{n}_{21}) - \vec{n}_{21} (K_f \cdot \vec{n}_{21})] \cdot \vec{t} = K_f \end{aligned}$$

$$\vec{D} \times \vec{n} = \vec{J}_m \rightarrow \vec{n}_{21} \times (\vec{D}_2 - \vec{D}_1) = \vec{J}_{m2}$$

例: 均匀磁化后的球形介质 $\vec{M} = -\frac{3}{2} \vec{H}_0$, 计算磁化电流分布.

$$\vec{K}_m = \vec{n} \times \vec{M} = (-\frac{3}{2}) \vec{n} \times \vec{H}_0 = -\frac{3}{2} H_0 \sin \theta (\vec{n} \times \vec{H}_0)$$

$\vec{H}_0$
↑

1, ^

§ 2.7 电磁场的能量、动量、角动量.

1. 场和电荷系统能量守恒定律.

能量密度 w , 能流密度 $\vec{S}$ Poynting 矢量.

$$-\oint_S d\vec{S} \cdot \vec{S} = \int_V \vec{J} \cdot \vec{v} dV + \frac{q}{\partial t} \int_V w dV$$

$$\vec{v} \cdot \vec{S} + \vec{J} \cdot \vec{v} + \frac{\partial w}{\partial t} = 0$$

2. 能量密度和能流密度

$$\vec{J} = \rho_f \vec{E} + \vec{J}_f \times \vec{B}$$

$$\vec{J} \cdot \vec{v} = \rho_f \vec{E} \cdot \vec{v} + (\rho_f \vec{v} \times \vec{B}) \cdot \vec{v} = \vec{J}_f \cdot \vec{E}$$

$$\hookrightarrow \vec{v} \cdot \vec{S} + \vec{J}_f \cdot \vec{E} + \frac{\partial w}{\partial t} = 0$$

1) 真空 $\vec{v} \times \vec{B} = \mu_0 \vec{J}_f + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$

$$\vec{E} \cdot (\vec{v} \times \vec{B}) = \mu_0 \vec{J}_f \cdot \vec{E} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \cdot \vec{E}$$

$$\begin{aligned} \vec{v} \cdot (\vec{E} \times \vec{B}) &= \vec{B} \cdot (\vec{v} \times \vec{E}) - \vec{E} \cdot (\vec{v} \times \vec{B}) \\ &= -\vec{B} \cdot \frac{\partial \vec{B}}{\partial t} - \mu_0 \vec{J}_f \cdot \vec{E} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \cdot \vec{E} \\ &= -\mu_0 \vec{J}_f \cdot \vec{E} - \frac{1}{2} \frac{\partial}{\partial t} [\mu_0 B^2 + \mu_0 \epsilon_0 E^2] \end{aligned}$$

$$\hookrightarrow \vec{S} = \vec{E} \times \vec{H} = \frac{1}{\mu_0} (\vec{E} \times \vec{B})$$

$$w = \frac{1}{2} (B^2 + \mu_0 H^2) = \frac{1}{2} (\epsilon_0 E^2 + \frac{B^2}{\mu_0})$$

2) 介质 (只有电磁场, $\vec{J}_f = 0$)

$$\vec{v} \times \vec{H} = \frac{\partial \vec{D}}{\partial t}, \quad \vec{E} \cdot (\vec{v} \times \vec{H}) = \vec{E} \cdot \frac{\partial \vec{D}}{\partial t} \quad \rightarrow \vec{S} = \vec{E} \times \vec{H}$$

$$\begin{aligned} \vec{v} \cdot (\vec{E} \times \vec{H}) &= \vec{H} \cdot (\vec{v} \times \vec{E}) - \vec{E} \cdot (\vec{v} \times \vec{H}) \\ &= -\vec{H} \cdot \frac{\partial \vec{B}}{\partial t} - \vec{E} \cdot \frac{\partial \vec{D}}{\partial t} \end{aligned} \quad w = \frac{1}{2} (\vec{H} \cdot \vec{B} + \vec{E} \cdot \vec{D})$$

3) 导电媒质

$$\vec{v} \times \vec{H} = \vec{J}_f + \frac{\partial \vec{D}}{\partial t}, \quad \vec{E} \cdot (\vec{v} \times \vec{H}) = \vec{E} \cdot \vec{J}_f + \vec{E} \cdot \frac{\partial \vec{D}}{\partial t}$$

$$\begin{aligned}\vec{G} \cdot (\vec{E} \times \vec{H}) &= \vec{H} \cdot (\vec{G} \times \vec{E}) - \vec{E} \cdot (\vec{G} \times \vec{H}) \\ &= -\vec{H} \cdot \frac{\partial \vec{B}}{\partial t} - \vec{E} \cdot \vec{J}_f - \vec{E} \cdot \frac{\partial \vec{D}}{\partial t}\end{aligned}$$

$$\hookrightarrow \vec{S} = \vec{E} \times \vec{H}$$

$$\omega = \frac{1}{2} (\vec{H} \cdot \vec{B} + \vec{E} \cdot \vec{D}) + \vec{J}_f \cdot \vec{E}$$

3. 电磁场的能量传输

例：同轴传输线内、外半径分别为 a, b ，导线电流为 I ，电压为 U 。

(1) 忽略电阻，计算 $\vec{S}$ 和传输功率。

$$\oint \vec{H} \cdot d\vec{l} = 2\pi r H = I \Rightarrow H = \frac{I}{2\pi r}$$

$$\frac{b}{I} \xrightarrow{a \leftarrow U}$$

设内壁线电荷密度为 τ

$$\int d\vec{S} \cdot \vec{E} = 2\pi r dl E = \frac{1}{\epsilon_0} \tau \cdot dl$$

$$\hookrightarrow E = \frac{\tau}{2\pi \epsilon_0 r}$$

$$\int_a^b E dr = \frac{\tau}{2\pi \epsilon_0} \ln\left(\frac{b}{a}\right) = U \Rightarrow \tau = \frac{2\pi \epsilon_0 U}{\ln\left(\frac{b}{a}\right)}$$

$$\vec{S} = \vec{E} \times \vec{H} = \frac{IU}{2\pi r \ln\left(\frac{b}{a}\right)} \vec{S}$$

提示：导线中电子运动。

电子运动： $n \sim 10^{20}$, $m \sim 10^{-31}$

$$J \sim 10^6 \text{ A/m}^2$$

$$E_K = n \cdot \frac{1}{2} m v^2 = \frac{1}{2} n m \left(\frac{I}{n e}\right)^2 \sim 10^{-18}$$

$$\text{传输功率: } P = \int_a^b \vec{S} \cdot 2\pi r dr = UI$$

(2) 考虑内导线有有限电导率，计算通过内导线进入导线内的能流密度

并证明进入导线的能流等于导线的损耗功率。

设电流在内导线表面 δ 范围内均匀分布。

$$E = \frac{\tau}{2\pi \epsilon_0 [a^2 - (a-\delta)^2]}, \quad H = \frac{I}{2\pi a}$$

$$S = E H = \frac{I^2}{2\pi^2 \epsilon_0 a [a^2 - (a-\delta)^2]}$$

$\Rightarrow$ 导线传输过程中的损耗是

$$dP = S \cdot 2\pi a dl = \frac{I^2}{2\pi a \epsilon_0 \delta} dl = I^2 dl \quad \text{电磁场} \frac{1}{2}$$

4. 电磁场的动量和动量密度。

$$\vec{f} = \rho_f \vec{E} + \vec{J}_f \times \vec{B}$$

$$= (\vec{G} \cdot \vec{D}) \vec{E} + (\vec{G} \times \vec{H} - \frac{\partial \vec{D}}{\partial t}) \times \vec{B}$$

$$= (\vec{G} \cdot \vec{B}) \vec{E} - \vec{B} \times (\vec{G} \times \vec{H} - \frac{\partial \vec{D}}{\partial t}) + (\vec{G} \cdot \vec{B}) \vec{H} - \vec{B} \times (\vec{G} \times \vec{E} + \frac{\partial \vec{B}}{\partial t})$$

$$= (\vec{v} \cdot \vec{B}) \vec{E} - \vec{B} \times (\vec{v} \times \vec{E}) + (\vec{v} \cdot \vec{E}) \vec{H} - \vec{E} \times (\vec{v} \times \vec{H}) - \frac{d}{dt} (\vec{r} \times \vec{B})$$

$$\vec{v}_E (\vec{B} \cdot \vec{E}) = (\vec{B} \cdot \vec{v}) \vec{E} + \vec{B} \times (\vec{v} \times \vec{E})$$

$$\vec{v}_H (\vec{E} \cdot \vec{H}) = (\vec{E} \cdot \vec{v}) \vec{H} + \vec{E} \times (\vec{v} \times \vec{H})$$

$$= (\vec{v} \cdot \vec{B}) \vec{E} - \vec{v}_E (\vec{B} \cdot \vec{E}) + (\vec{B} \cdot \vec{v}) \vec{E} + (\vec{v} \cdot \vec{E}) \vec{H} - \vec{v}_H (\vec{E} \cdot \vec{H}) - \frac{d}{dt} (\vec{r} \times \vec{B})$$

$$= \vec{v} \cdot (\vec{B} \vec{E}) - \vec{v}_E (\vec{B} \cdot \vec{E}) + \vec{v} (\vec{B} \cdot \vec{H}) - \vec{v}_H (\vec{E} \cdot \vec{H}) - \frac{d}{dt} (\vec{r} \times \vec{B})$$

$$\vec{v}_E (\vec{B} \cdot \vec{E})_j = e_j \partial_j (\vec{B} \cdot \vec{E})$$

$$= \vec{B} \cdot e_j \partial_j \vec{E} = E_i E_i e_j \partial_j E_i = E \vec{E} \cdot (\vec{v} \times \vec{E})$$

$$\vec{v} (\vec{B} \cdot \vec{H}) = e_j \partial_j (\vec{B} \cdot \vec{H})$$

$$= e_j \partial_j (E E_i H_i) = E_i E_i e_j \partial_j E + 2 E E_i e_j \partial_j E_i$$

$$= \vec{E}^2 \vec{v} E + 2 E \vec{E} \cdot (\vec{v} \times \vec{E})$$

$$\hookrightarrow \vec{v}_E (\vec{B} \cdot \vec{E}) = \frac{1}{2} [\vec{v} (\vec{B} \cdot \vec{E}) - \vec{E}^2 \vec{v} E]$$

$$\text{同理: } \vec{v}_H (\vec{E} \cdot \vec{H}) = \frac{1}{2} [\vec{v} (\vec{E} \cdot \vec{H}) - \vec{H}^2 \vec{v} H]$$

$$\vec{f}_m = \rho_f \vec{E} + \vec{J}_f \times \vec{B} - \frac{1}{2} \vec{E}^2 \vec{v} E - \frac{1}{2} \vec{H}^2 \vec{v} H$$

$$= \vec{v} \cdot (\vec{B} \vec{E}) + \vec{v} \cdot (\vec{E} \vec{H}) - \frac{1}{2} [\vec{v} (\vec{B} \cdot \vec{E}) + \vec{v} (\vec{E} \cdot \vec{H})] - \frac{d}{dt} (\vec{r} \times \vec{B})$$

$$= \vec{v} \cdot \left[\vec{B} \vec{E} + \vec{E} \vec{H} - \frac{1}{2} (\vec{B} \cdot \vec{E} + \vec{E} \cdot \vec{H}) \vec{v} \right] - \frac{d}{dt} (\vec{r} \times \vec{B})$$

$$= -\vec{v} \cdot \vec{T} - \frac{d}{dt} g_m$$

$$\text{动量流密度 } \vec{T} = \vec{B} \vec{E} + \vec{E} \vec{H} - \frac{1}{2} (\vec{B} \cdot \vec{E} + \vec{E} \cdot \vec{H}) \vec{v}$$

$$\text{能量密度 } g_m = \vec{B} \times \vec{E} = \epsilon_0 \vec{E} \times \vec{H} = g_m \vec{v}$$

$$\Rightarrow \vec{f}_m + \frac{d}{dt} g_m = -\vec{v} \cdot \vec{T}$$

例: 质量为 m , 电荷为 q 的点电荷在静磁场 $\vec{B}$ 中速度 $\vec{v}$ 运动, 计算电磁动量和总动量.

$$\vec{G} = \epsilon_0 \int_V \vec{E} \times \vec{B} dV = \epsilon_0 \int_V \vec{E} \times (\vec{v} \times \vec{B}) dV$$

$$\vec{v} (\vec{B} \cdot \vec{E}) = (\vec{E} \cdot \vec{v}) \vec{B} + (\vec{B} \cdot \vec{v}) \vec{E} + \vec{E} \times (\vec{v} \times \vec{B}) + \vec{B} \times (\vec{v} \times \vec{E})$$

$$= \epsilon_0 \int_V [\vec{v} (\vec{B} \cdot \vec{E}) - (\vec{E} \cdot \vec{v}) \vec{B} - (\vec{B} \cdot \vec{v}) \vec{E} - \vec{B} \times (\vec{v} \times \vec{E})] dV$$

$$\vec{v} (\vec{B} \cdot \vec{E}) = (\vec{v} \cdot \vec{B}) \vec{E} + (\vec{B} \cdot \vec{v}) \vec{E} = (\vec{B} \cdot \vec{v}) \vec{E} \quad \hookrightarrow \text{类似静电场}$$

$$\vec{v} (\vec{E} \cdot \vec{B}) = (\vec{v} \cdot \vec{E}) \vec{B} + (\vec{E} \cdot \vec{v}) \vec{B} = \frac{q}{\epsilon_0} \delta(\vec{r} - \vec{v}t) \vec{B} + (\vec{E} \cdot \vec{v}) \vec{B}$$

$$= \epsilon_0 \int_V [\vec{v} (\vec{B} \cdot \vec{E}) - \vec{v} (\vec{E} \cdot \vec{B}) - \vec{v} (\vec{E} \cdot \vec{B}) + \frac{q}{\epsilon_0} \delta(\vec{r} - \vec{v}t) \vec{B}] dV$$

$$= \epsilon_0 \int_V \nabla \cdot (\vec{A} \cdot \vec{E}) dV - \vec{A} \cdot \vec{E} - \vec{E} \cdot \vec{A} + q \int_V \delta(R - vt) A dV$$

$$= \epsilon_0 \oint_{\infty} [\vec{A} \cdot \vec{E}] dS - \vec{A} \cdot \vec{E} - \vec{E} \cdot \vec{A} + q \vec{A} = 0$$

$$= q \vec{A} \quad \text{对有限区域的电荷/电流分布, } R \rightarrow \infty, \vec{A}, \vec{E} \rightarrow 0 \text{ 的速度} > \frac{1}{R}$$

$$\text{总动量 (正则动量)} \quad \vec{p} = m\vec{v} + q\vec{A}$$

例: 计算空心同轴传输线单位长度内的电磁场总动量。

$$\vec{g}_m = \mu_0 \epsilon_0 \vec{S} = \frac{\mu_0 I^2}{4\pi^2 r^2} \vec{e}_r$$

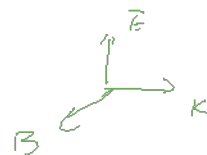
$$\vec{G}_m = \int_0^b 2\pi r dr \vec{g}_m = \frac{\mu_0 I^2}{2\pi} \int_0^b \frac{1}{r} dr \vec{e}_r = \frac{\mu_0 I^2}{2\pi} \ln\left(\frac{b}{a}\right) \vec{e}_r$$

例: 计算平面电磁波的动量密度。

$$\vec{E} = \vec{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)}$$

$$\frac{\partial \vec{E}}{\partial t} = -\nabla \times \vec{Z} = -i\vec{k} \times \vec{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)}$$

$$\hookrightarrow \vec{B} = \frac{\vec{k}}{\omega} \times \vec{E}$$



$$\vec{S} = \frac{1}{\mu_0} \langle \vec{E} \times \vec{B} \rangle = \frac{1}{\mu_0 \omega} \left[\vec{E} \times (\vec{k} \times \vec{E}) \right]$$

$$= \frac{k E_0^2}{\mu_0 \omega} \vec{k}$$

$$k = \frac{2\pi}{\lambda}$$

$$\frac{k}{\omega} = \frac{2\pi}{\omega \lambda} = \frac{2\pi}{2\pi \lambda} = \frac{1}{\lambda} = \frac{1}{c}$$

$$\vec{g} = \mu_0 \epsilon_0 \vec{S} = \frac{\epsilon_0 k E_0^2}{\omega} \vec{k} =$$

$$= \frac{1}{c^2} \mu_0$$

例: 计算平面电磁波的动量流密度。

1. 计算 (1) 用 \$E\$ 和 \$B\$ 表示 \$\vec{f}_m\$ 和 \$\vec{g}_m\$ 的表达式

5. 电磁场的动量

$$\vec{R} \times \vec{f}_m \, dV + \vec{R} \times \frac{\partial \vec{g}_m}{\partial t} \, dV = - \vec{R} \times (\nabla \cdot \vec{T} \, dV)$$

$$\vec{M}_{mech} := \vec{R} \times \vec{f}_m = \frac{d\vec{L}_{mech}}{dt}$$

$$\frac{d\vec{L}_{mech}}{dt} + \frac{d}{dt} \int_V (\vec{R} \times \vec{g}_m) \, dV = - \int_V \vec{R} \times (\nabla \cdot \vec{T}) \, dV$$

$$\vec{R} \times (\nabla \cdot \vec{T}) = - \nabla \cdot (\vec{T} \times \vec{R})$$

$$= R_i \vec{e}_i \times (\vec{e}_j \partial_j \cdot T_{km} \vec{e}_k \vec{e}_m) = R_i \vec{e}_i \times (\partial_j T_{km} \delta_{ij} \vec{e}_m)$$

$$= R_i \vec{e}_i \times (\partial_j T_{jm} \vec{e}_m) = \epsilon_{nim} \vec{e}_n R_i \partial_j T_{jm}$$

$$= \epsilon_{nim} \vec{e}_n [\partial_j (R_i T_{jm}) - T_{jm} \partial_j R_i] \quad \vec{T} \text{ 是反对称张量。}$$

$$= \epsilon_{nim} \vec{e}_n [\partial_j (R_i T_{jm}) - T_{im}] \quad \epsilon_{nim} \vec{e}_n \underline{T_{im}} = \epsilon_{nim} \vec{e}_n T_{mi} = - \epsilon_{nim} \vec{e}_n T_{im} = 0$$

$$= \epsilon_{nim} \vec{e}_n \partial_j (R_i T_{jm}) = - \epsilon_{nmi} \vec{e}_n \partial_j (T_{jm} R_i)$$

$$= - \nabla \cdot (\vec{T} \times \vec{R})$$

$$\begin{aligned} \frac{d\vec{L}_{mech}}{dt} + \frac{d}{dt} \int_V (\vec{R} \times \vec{g}_m) \, dV &= - \int_V \nabla \cdot (\vec{T} \times \vec{R}) \, dV \\ &= - \oint_S (-\vec{T} \times \vec{R}) \cdot d\vec{S} \end{aligned}$$

$$\text{电磁场动量密度 } \vec{L}_{em} = \vec{R} \times \vec{g}_m$$

$$\text{电磁场动量流密度 } \vec{M}_{em} = - \vec{T} \times \vec{R}$$

$$\begin{aligned} \frac{d}{dt} \langle \vec{L}_{mech} + \vec{L}_{em} \rangle &= - \oint_S \vec{M}_{em} \cdot d\vec{S} \\ &= 0 \quad (S \rightarrow \infty) \end{aligned}$$